EECS 182 – Homework 1 Warm-Up Worksheet Solutions

1 Vector and Matrix Calculus Refresher

Let $\boldsymbol{x}, \boldsymbol{c} \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$.

1.1 $f(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{c}$

(a) $\dfrac{\partial f}{\partial \boldsymbol{x}}$ is a row vector of length n.

(b) Write $f(\boldsymbol{x}) = \sum_{i=1}^n x_i c_i$. Then

$$\frac{\partial f}{\partial x_i} = c_i \quad \Rightarrow \quad \frac{\partial f}{\partial \boldsymbol{x}} = \boldsymbol{c}^\top.$$

1.2 $f(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{x}$

(a) $f(\boldsymbol{x}) = \sum_{i=1}^n x_i^2$.

(b) Then

$$\frac{\partial f}{\partial x_i} = 2x_i \quad \Rightarrow \quad \frac{\partial f}{\partial \boldsymbol{x}} = 2\boldsymbol{x}^\top.$$

1.3 $\boldsymbol{f}(\boldsymbol{x}) = A\boldsymbol{x}$

(a) $\dfrac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}$ is an $n \times n$ Jacobian matrix.

(b) Since $\boldsymbol{f}(\boldsymbol{x}) = A\boldsymbol{x}$ is linear, the Jacobian is just A.

1.4 $f(\boldsymbol{x}) = \boldsymbol{x}^\top A \boldsymbol{x}$

(a) Writing $A = (a_{ij})$,

$$f(\boldsymbol{x}) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j.$$

(b) Fix i and differentiate:

$$\frac{\partial f}{\partial x_i} = \sum_j a_{ij} x_j + \sum_j a_{ji} x_j = (A\boldsymbol{x})_i + (A^\top \boldsymbol{x})_i.$$

(c) Stacking over i,

$$\frac{\partial f}{\partial \boldsymbol{x}} = \boldsymbol{x}^\top (A + A^\top).$$

(d) If A is symmetric ($A = A^\top$), then

$$\frac{\partial f}{\partial \boldsymbol{x}} = 2\boldsymbol{x}^\top A.$$

2 SVD, Least Squares, and Minimum-Norm Solutions

Let $X \in \mathbb{R}^{m\times n}$, $\boldsymbol{y} \in \mathbb{R}^m$, and $X = U\Sigma V^\top$.

2.1 Overdetermined case ($m > n$, full column rank)

(a) The normal equations give

$$X^\top X \boldsymbol{w}^\star = X^\top \boldsymbol{y} \quad \Rightarrow \quad \boldsymbol{w}^\star = (X^\top X)^{-1} X^\top \boldsymbol{y}.$$

(b) Using $X = U\Sigma V^\top$,

$$X^\top X = V\Sigma^\top U^\top U \Sigma V^\top = V(\Sigma^\top \Sigma)V^\top.$$

Hence

$$(X^\top X)^{-1} = V(\Sigma^\top \Sigma)^{-1} V^\top.$$

Therefore

$$\begin{aligned} \boldsymbol{w}^\star &= (X^\top X)^{-1} X^\top \boldsymbol{y} \\ &= V(\Sigma^\top \Sigma)^{-1} V^\top (V\Sigma^\top U^\top)\boldsymbol{y} \\ &= V(\Sigma^\top \Sigma)^{-1} \Sigma^\top U^\top \boldsymbol{y}. \end{aligned}$$

Since $\Sigma^\top\Sigma$ is diagonal with entries σ_i^2, $(\Sigma^\top\Sigma)^{-1}\Sigma^\top$ is diagonal with entries $1/\sigma_i$. This is precisely Σ^+, so

$$\boldsymbol{w}^\star = V\Sigma^+ U^\top \boldsymbol{y}.$$

2.2 Underdetermined case ($m < n$, full row rank)

(a) We solve

$$\min_{\boldsymbol{w}} \|\boldsymbol{w}\|_2^2 \quad \text{subject to} \quad X\boldsymbol{w} = \boldsymbol{y}.$$

(b) Form the Lagrangian

$$\mathcal{L}(\boldsymbol{w}, \boldsymbol{\lambda}) = \|\boldsymbol{w}\|_2^2 + \boldsymbol{\lambda}^\top (X\boldsymbol{w} - \boldsymbol{y}).$$

Setting the derivative with respect to $\boldsymbol{w}$ to zero:

$$2\boldsymbol{w} + X^\top \boldsymbol{\lambda} = 0 \quad \Rightarrow \quad \boldsymbol{w} = -\frac{1}{2} X^\top \boldsymbol{\lambda}.$$

Substitute into $X\boldsymbol{w} = \boldsymbol{y}$:

$$X\left(-\frac{1}{2}X^\top \boldsymbol{\lambda}\right) = \boldsymbol{y} \quad \Rightarrow \quad -\frac{1}{2} X X^\top \boldsymbol{\lambda} = \boldsymbol{y}.$$

Thus $\boldsymbol{\lambda} = -2(XX^\top)^{-1}\boldsymbol{y}$, and

$$\boldsymbol{w}_{\min} = X^\top (XX^\top)^{-1} \boldsymbol{y}.$$

(c) Using the SVD, $XX^\top = U\Sigma\Sigma^\top U^\top$, so

$$(XX^\top)^{-1} = U(\Sigma\Sigma^\top)^{-1}U^\top.$$

Then

$$\begin{aligned}\boldsymbol{w}_{\min} &= X^\top(XX^\top)^{-1}\boldsymbol{y} \\ &= V\Sigma^\top U^\top\, U(\Sigma\Sigma^\top)^{-1}U^\top\boldsymbol{y} \\ &= V\Sigma^\top(\Sigma\Sigma^\top)^{-1}U^\top\boldsymbol{y}.\end{aligned}$$

The diagonal matrix $\Sigma^\top(\Sigma\Sigma^\top)^{-1}$ has entries $1/\sigma_i$, so it is Σ^+. Thus

$$\boldsymbol{w}_{\min} = V\Sigma^+U^\top\boldsymbol{y}.$$

2.3 Concept check

(a) A *least-squares* solution minimizes the residual norm $\|X\boldsymbol{w}-\boldsymbol{y}\|_2$. A *minimum-norm* solution is the solution among all exact fits $X\boldsymbol{w}=\boldsymbol{y}$ that has the smallest $\|\boldsymbol{w}\|_2$.

(b) With full column rank ($m > n$), the least-squares solution is unique. With full row rank ($m < n$), the minimum-norm solution is unique.

3 Ridge Regression Basics

Let $X \in \mathbb{R}^{n\times d}$, $\boldsymbol{y} \in \mathbb{R}^n$, $\lambda > 0$, and

$$L(\boldsymbol{w}) = \|\boldsymbol{y} - X\boldsymbol{w}\|_2^2 + \lambda\|\boldsymbol{w}\|_2^2.$$

3.1 Closed-form solution

(a) Expand:

$$L(\boldsymbol{w}) = (\boldsymbol{y} - X\boldsymbol{w})^\top(\boldsymbol{y} - X\boldsymbol{w}) + \lambda\boldsymbol{w}^\top\boldsymbol{w}.$$

The gradient is

$$\frac{\partial L}{\partial \boldsymbol{w}} = -2X^\top(\boldsymbol{y} - X\boldsymbol{w}) + 2\lambda\boldsymbol{w}.$$

(b) Setting this to zero:

$$-2X^\top(\boldsymbol{y} - X\boldsymbol{w}) + 2\lambda\boldsymbol{w} = 0$$

gives

$$X^\top X\boldsymbol{w} - X^\top\boldsymbol{y} + \lambda\boldsymbol{w} = 0 \quad\Rightarrow\quad (X^\top X + \lambda I)\boldsymbol{w} = X^\top\boldsymbol{y}.$$

So

$$\boldsymbol{w}_{\text{ridge}} = (X^\top X + \lambda I)^{-1}X^\top\boldsymbol{y}.$$

3.2 SVD perspective and shrinkage

Let $X = U\Sigma V^\top$.

(a) Then $X^\top X = V\Sigma^\top \Sigma V^\top$, so

$$\boldsymbol{w}_{\text{ridge}} = V(\Sigma^\top\Sigma + \lambda I)^{-1}\Sigma^\top U^\top \boldsymbol{y}.$$

The only non-identity diagonal matrix is $(\Sigma^\top\Sigma + \lambda I)^{-1}\Sigma^\top$, which acts diagonally on singular directions.

(b) Suppose $U^\top \boldsymbol{y}$ has components α_i along singular vectors. In the V-basis, the coefficient along direction i is

$$\frac{\sigma_i}{\sigma_i^2 + \lambda}\,\alpha_i.$$

Equivalently, relative to the OLS coefficient (which scales like α_i/σ_i), ridge multiplies by

$$\frac{\sigma_i^2}{\sigma_i^2 + \lambda}.$$

(c) If σ_i is very small, $\frac{\sigma_i^2}{\sigma_i^2+\lambda} \approx 0$, so that direction is heavily shrunk. If σ_i is very large, the factor is close to 1, so that direction is only mildly shrunk.

3.3 Interpretation

(a) Ridge is called "shrinkage" because it shrinks coefficients toward zero, especially along directions that X does not constrain well (those with small σ_i).

(b) As $\lambda \to 0$, $\boldsymbol{w}_{\text{ridge}} \to (X^\top X)^{-1}X^\top \boldsymbol{y}$, the ordinary least-squares solution (assuming $X^\top X$ is invertible).

(c) As $\lambda \to \infty$, the penalty dominates and $\boldsymbol{w}_{\text{ridge}} \to \mathbf{0}$.

4 Scalar Gradient Descent Stability

Consider $L(w) = (y - \sigma w)^2$ with $\sigma > 0$ and gradient descent

$$w_{t+1} = w_t - \eta\,\frac{dL}{dw}\Big|_{w_t}.$$

4.1 Derive the update

(a) Expand:

$$L(w) = (y - \sigma w)^2 = (\sigma w - y)^2.$$

Then

$$\frac{dL}{dw} = 2(\sigma w - y)\,\sigma = 2\sigma(\sigma w - y).$$

(b) The update is

$$\begin{aligned} w_{t+1} &= w_t - \eta \cdot 2\sigma(\sigma w_t - y) \\ &= w_t - 2\eta\sigma^2 w_t + 2\eta\sigma y \\ &= (1 - 2\eta\sigma^2)w_t + 2\eta\sigma y. \end{aligned}$$

(c) The optimal $w^\star$ solves $\frac{dL}{dw} = 0$, i.e. $\sigma w^\star - y = 0$, so

$$w^\star = \frac{y}{\sigma}.$$

4.2 Error dynamics

Let $e_t = w_t - w^\star$.

(a) Write

$$w_t = e_t + w^\star.$$

Then

$$w_{t+1} = (1 - 2\eta\sigma^2)(e_t + w^\star) + 2\eta\sigma y.$$

Using $w^\star = y/\sigma$,

$$(1 - 2\eta\sigma^2)w^\star + 2\eta\sigma y = (1 - 2\eta\sigma^2)\frac{y}{\sigma} + 2\eta\sigma y = \frac{y}{\sigma} = w^\star.$$

Hence

$$w_{t+1} - w^\star = (1 - 2\eta\sigma^2)e_t.$$

(b) So

$$e_{t+1} = (1 - 2\eta\sigma^2)e_t,$$

and by iteration,

$$e_t = (1 - 2\eta\sigma^2)^t e_0.$$

The common ratio is

$$r(\eta, \sigma) = 1 - 2\eta\sigma^2.$$

4.3 Stability region

(a) We need $|r| < 1$ for convergence, i.e.

$$|1 - 2\eta\sigma^2| < 1.$$

(b) This inequality is equivalent to

$$-1 < 1 - 2\eta\sigma^2 < 1 \quad \Rightarrow \quad 0 < 2\eta\sigma^2 < 2 \quad \Rightarrow \quad 0 < \eta < \frac{1}{\sigma^2}.$$

4.4 Convergence speed

(a) If $|e_t| \le \alpha^t |e_0|$ with $\alpha \in (0, 1)$, then to ensure $|e_t| \le \varepsilon |e_0|$ we need

$$\alpha^t \le \varepsilon \quad \Rightarrow \quad t \ge \frac{\log(\varepsilon)}{\log(\alpha)}.$$

(Note $\log(\alpha) < 0$.)

(b) Here $\alpha = |1 - 2\eta\sigma^2|$, so

$$t \;\ge\; \frac{\log(\varepsilon)}{\log|1 - 2\eta\sigma^2|}.$$

5 Two-Dimensional Diagonal System and Condition Number

The system is

$$\begin{bmatrix} \sigma_L & 0 \\ 0 & \sigma_S \end{bmatrix} \begin{bmatrix} w^{[1]} \\ w^{[2]} \end{bmatrix} = \begin{bmatrix} y^{[1]} \\ y^{[2]} \end{bmatrix}, \quad \sigma_L \gg \sigma_S > 0.$$

5.1 Mode-wise behavior

(a) The squared loss decomposes as a sum of two independent 1D losses, one involving $(\sigma_L, w^{[1]}, y^{[1]})$ and one involving $(\sigma_S, w^{[2]}, y^{[2]})$. Thus each coordinate follows the scalar gradient descent dynamics from Section 4 with its own σ.

(b) For $k \in \{1, 2\}$ with $\sigma_k \in \{\sigma_L, \sigma_S\}$, the error satisfies

$$e_{t+1}^{(k)} = (1 - 2\eta\sigma_k^2)\, e_t^{(k)}.$$

Hence

$$r_L(\eta) = 1 - 2\eta\sigma_L^2, \qquad r_S(\eta) = 1 - 2\eta\sigma_S^2.$$

5.2 Stability and limiting factor

(a) We require $|1 - 2\eta\sigma_L^2| < 1$ and $|1 - 2\eta\sigma_S^2| < 1$. Since $\sigma_L > \sigma_S$, the first condition is stricter. Hence we must choose

$$0 < \eta < \frac{1}{\sigma_L^2}.$$

(b) The larger singular value σ_L limits how large η can be while still ensuring stability for both coordinates.

5.3 Relative convergence speeds

(a) For a fixed η in the stable range, the factor $|1 - 2\eta\sigma^2|$ is smaller when σ is larger (up to the stability limit). Thus the component associated with σ_L converges faster than the one associated with σ_S.

(b) The condition number $\kappa = \sigma_L^2/\sigma_S^2$ measures how stretched the problem is. Large κ forces η to be small (to keep the large-σ mode stable), which makes the small-σ mode converge very slowly, slowing overall convergence.

6 Gradient Descent and SGD on Simple Models

6.1 Constant model: $\hat{y}_i = \theta$

We have

$$L(\theta) = \frac{1}{N} \sum_{i=1}^{N} (y_i - \theta)^2.$$

(a) Differentiate:

$$\frac{\partial L}{\partial \theta} = \frac{1}{N} \sum_i 2(y_i - \theta)(-1) = \frac{2}{N} \sum_i (\theta - y_i).$$

(b) Set to zero:

$$\frac{2}{N}\sum_i(\theta - y_i) = 0 \quad \Rightarrow \quad \theta\sum_i 1 = \sum_i y_i \quad \Rightarrow \quad \theta^\star = \frac{1}{N}\sum_{i=1}^N y_i.$$

(c) Full-batch GD:

$$\theta \leftarrow \theta - \alpha \cdot \frac{2}{N}\sum_{i=1}^N(\theta - y_i).$$

(d) Single-sample SGD (for a sampled i):

$$\theta \leftarrow \theta - \alpha \cdot 2(\theta - y_i).$$

(e) Each point contributes a term of the same functional form $(\theta - y_i)$, with no x_i scaling, so every data point has an equal "vote" in the gradient.

6.2 Linear model: $\hat{y}_i = \theta x_i$

Now

$$L(\theta) = \frac{1}{N}\sum_{i=1}^N (y_i - \theta x_i)^2.$$

(a) Differentiate:

$$\frac{\partial L}{\partial \theta} = \frac{1}{N}\sum_i 2(y_i - \theta x_i)(-x_i) = \frac{2}{N}\sum_i(\theta x_i^2 - x_i y_i).$$

(b) Set to zero:

$$\frac{2}{N}\sum_i(\theta x_i^2 - x_i y_i) = 0 \quad \Rightarrow \quad \theta\sum_i x_i^2 = \sum_i x_i y_i,$$

so

$$\theta^\star = \frac{\sum_i x_i y_i}{\sum_i x_i^2}.$$

(c) Full-batch GD update:

$$\theta \leftarrow \theta - \alpha \cdot \frac{2}{N}\sum_{i=1}^N(\theta x_i - y_i)x_i.$$

(d) Single-sample SGD (for a sampled i):

$$\theta \leftarrow \theta - \alpha \cdot 2(\theta x_i - y_i)x_i.$$

(e) No: points do not contribute equally in magnitude. Their contributions are scaled by $|x_i|$ (or x_i^2 in the sum), so points with larger $|x_i|$ have larger influence on the gradient.

7 Noise as Regularization

7.1 One-dimensional sanity check

Let $\tilde{x} = x + N$, $N \sim \mathcal{N}(0, \sigma^2)$, and consider $(\tilde{x}w - y)^2$.

(a) Expand:

$$\begin{aligned}(\tilde{x}w - y)^2 &= ((x + N)w - y)^2 \\ &= (xw - y + Nw)^2 \\ &= (xw - y)^2 + 2(xw - y)(Nw) + (Nw)^2.\end{aligned}$$

Taking expectation:

$$\mathbb{E}[(xw - y)^2] = (xw - y)^2 \quad \text{(no randomness)},$$

and $\mathbb{E}[N] = 0$ so the cross term vanishes. Also,

$$\mathbb{E}[(Nw)^2] = w^2 \mathbb{E}[N^2] = w^2 \sigma^2.$$

Thus

$$\mathbb{E}[(\tilde{x}w - y)^2] = (xw - y)^2 + \sigma^2 w^2.$$

(b) The “something” multiplying w^2 is σ^2.

7.2 Matrix case

Let X be $m \times n$ with rows $\boldsymbol{x}_i^\top$, and $\tilde{X} = X + N$ with rows $\tilde{\boldsymbol{x}}_i^\top = \boldsymbol{x}_i^\top + \boldsymbol{N}_i^\top$.

(a) We can write

$$\|\tilde{X}\boldsymbol{w} - \boldsymbol{y}\|_2^2 = (\tilde{X}\boldsymbol{w} - \boldsymbol{y})^\top (\tilde{X}\boldsymbol{w} - \boldsymbol{y}).$$

(b) Substitute $\tilde{X} = X + N$:

$$\tilde{X}\boldsymbol{w} - \boldsymbol{y} = (X\boldsymbol{w} - \boldsymbol{y}) + N\boldsymbol{w},$$

so

$$\|\tilde{X}\boldsymbol{w} - \boldsymbol{y}\|_2^2 = \|X\boldsymbol{w} - \boldsymbol{y}\|_2^2 + 2(X\boldsymbol{w} - \boldsymbol{y})^\top N\boldsymbol{w} + \|N\boldsymbol{w}\|_2^2.$$

Taking expectation, the middle term vanishes since each $\boldsymbol{N}_i$ has mean $\mathbf{0}$ and is independent of $(X, \boldsymbol{y})$. For the last term:

$$\|N\boldsymbol{w}\|_2^2 = \sum_{i=1}^{m} (\boldsymbol{N}_i^\top \boldsymbol{w})^2.$$

Each $\boldsymbol{N}_i$ has covariance $\sigma^2 I$, so

$$\mathbb{E}[(\boldsymbol{N}_i^\top \boldsymbol{w})^2] = \boldsymbol{w}^\top \mathbb{E}[\boldsymbol{N}_i \boldsymbol{N}_i^\top] \boldsymbol{w} = \boldsymbol{w}^\top (\sigma^2 I) \boldsymbol{w} = \sigma^2 \|\boldsymbol{w}\|_2^2.$$

Summing over i:

$$\mathbb{E}[\|N\boldsymbol{w}\|_2^2] = m\sigma^2 \|\boldsymbol{w}\|_2^2.$$

Thus

$$\mathbb{E}[\|\tilde{X}\boldsymbol{w} - \boldsymbol{y}\|_2^2] = \|X\boldsymbol{w} - \boldsymbol{y}\|_2^2 + m\sigma^2 \|\boldsymbol{w}\|_2^2.$$

(c) Hence $\lambda = m\sigma^2$.

7.3 Concept

(a) Adding isotropic Gaussian noise to inputs makes the expected loss include an extra term proportional to $\|\boldsymbol{w}\|_2^2$, which is exactly ℓ_2 regularization. Intuitively, large weights make predictions very sensitive to input noise, and the optimization penalizes such sensitivity.

(b) As σ^2 increases, the effective λ increases, so regularization becomes stronger and the solution shrinks more toward zero.

8 General Tikhonov / Weighted Least Squares

8.1 1D toy version

Consider

$$f(x) = (ax - b)^2 + \alpha(x - c)^2, \quad a, b, c, \alpha > 0.$$

(a) Expand:

$$\begin{aligned} f(x) &= (a^2x^2 - 2abx + b^2) + (\alpha x^2 - 2\alpha cx + \alpha c^2) \\ &= (a^2 + \alpha)x^2 - 2(ab + \alpha c)x + (b^2 + \alpha c^2). \end{aligned}$$

Then

$$\frac{df}{dx} = 2(a^2 + \alpha)x - 2(ab + \alpha c).$$

(b) Set derivative to zero:

$$2(a^2 + \alpha)x - 2(ab + \alpha c) = 0 \quad \Rightarrow \quad x^\star = \frac{ab + \alpha c}{a^2 + \alpha}.$$

(c) Rewrite:

$$x^\star = \frac{a^2}{a^2 + \alpha} \cdot \frac{b}{a} + \frac{\alpha}{a^2 + \alpha} \cdot c.$$

So $x^\star$ is a weighted average of b/a and c with weights proportional to a^2 and α.

8.2 Matrix expansion

For

$$\min_{\boldsymbol{x}} \|W_1(A\boldsymbol{x} - \boldsymbol{b})\|_2^2 + \|W_2(\boldsymbol{x} - \boldsymbol{c})\|_2^2,$$

write

$$(A\boldsymbol{x} - \boldsymbol{b})^\top W_1^\top W_1(A\boldsymbol{x} - \boldsymbol{b}) + (\boldsymbol{x} - \boldsymbol{c})^\top W_2^\top W_2(\boldsymbol{x} - \boldsymbol{c}).$$

Taking the gradient with respect to $\boldsymbol{x}$:

$$\frac{\partial}{\partial \boldsymbol{x}} = 2A^\top W_1^\top W_1(A\boldsymbol{x} - \boldsymbol{b}) + 2W_2^\top W_2(\boldsymbol{x} - \boldsymbol{c}).$$

Setting to zero:

$$A^\top W_1^\top W_1(A\boldsymbol{x} - \boldsymbol{b}) + W_2^\top W_2(\boldsymbol{x} - \boldsymbol{c}) = 0.$$

Rearranging:

$$(A^\top W_1^\top W_1 A + W_2^\top W_2)\boldsymbol{x} = A^\top W_1^\top W_1 \boldsymbol{b} + W_2^\top W_2 \boldsymbol{c}.$$

Thus

$$\boldsymbol{x}^\star = (A^\top W_1^\top W_1 A + W_2^\top W_2)^{-1}\big(A^\top W_1^\top W_1 \boldsymbol{b} + W_2^\top W_2 \boldsymbol{c}\big).$$

8.3 Stacking trick

(a) Define

$$C = \begin{bmatrix} W_1 A \\ W_2 I \end{bmatrix}, \quad \boldsymbol{d} = \begin{bmatrix} W_1 \boldsymbol{b} \\ W_2 \boldsymbol{c} \end{bmatrix}.$$

Then

$$C\boldsymbol{x} - \boldsymbol{d} = \begin{bmatrix} W_1(A\boldsymbol{x} - \boldsymbol{b}) \\ W_2(\boldsymbol{x} - \boldsymbol{c}) \end{bmatrix},$$

and hence

$$\|C\boldsymbol{x} - \boldsymbol{d}\|_2^2 = \|W_1(A\boldsymbol{x} - \boldsymbol{b})\|_2^2 + \|W_2(\boldsymbol{x} - \boldsymbol{c})\|_2^2.$$

(b) Now

$$C^\top C = A^\top W_1^\top W_1 A + W_2^\top W_2, \quad C^\top \boldsymbol{d} = A^\top W_1^\top W_1 \boldsymbol{b} + W_2^\top W_2 \boldsymbol{c}.$$

The OLS solution $\boldsymbol{x}^\star = (C^\top C)^{-1} C^\top \boldsymbol{d}$ matches the formula in 8.2.

(c) For ridge regression set $W_1 = I$, $W_2 = \sqrt{\lambda} I$, $\boldsymbol{c} = \boldsymbol{0}$, and rename $A = X$, $\boldsymbol{b} = \boldsymbol{y}$. Then

$$A^\top W_1^\top W_1 A + W_2^\top W_2 = X^\top X + \lambda I,$$

and

$$A^\top W_1^\top W_1 \boldsymbol{b} + W_2^\top W_2 \boldsymbol{c} = X^\top \boldsymbol{y} + 0,$$

so

$$\boldsymbol{x}^\star = (X^\top X + \lambda I)^{-1} X^\top \boldsymbol{y},$$

which is exactly ridge regression.

9 Gaussian MAP and Ridge Regression

Let $W \in \mathbb{R}^d$, $Y \in \mathbb{R}^n$ with

$$W \sim \mathcal{N}(0, I_d), \quad N \sim \mathcal{N}(0, I_n), \quad Y = XW + \sqrt{\lambda}\, N, \quad N \perp W.$$

9.1 Covariances

(a) By definition,

$$\Sigma_{WW} = \mathbb{E}[WW^\top] = I_d.$$

(b) Expand

$$Y = XW + \sqrt{\lambda}\, N.$$

Then

$$\begin{aligned} \Sigma_{YY} &= \mathbb{E}[YY^\top] = \mathbb{E}[(XW + \sqrt{\lambda} N)(XW + \sqrt{\lambda} N)^\top] \\ &= \mathbb{E}[XWW^\top X^\top] + \lambda \mathbb{E}[NN^\top] + \mathbb{E}[XWN^\top]\sqrt{\lambda} + \sqrt{\lambda}\,\mathbb{E}[NW^\top]X^\top. \end{aligned}$$

Independence and zero-mean imply the cross terms vanish, so

$$\Sigma_{YY} = X\mathbb{E}[WW^\top]X^\top + \lambda I_n = XX^\top + \lambda I.$$

(c) For Σ_{WY}:

$$\begin{aligned} \Sigma_{WY} &= \mathbb{E}[WY^\top] = \mathbb{E}[W(XW + \sqrt{\lambda} N)^\top] \\ &= \mathbb{E}[WW^\top X^\top] + \sqrt{\lambda}\mathbb{E}[WN^\top] = X^\top. \end{aligned}$$

9.2 Conditional expectation and ridge form

For jointly zero-mean Gaussian (W, Y),

$$\mathbb{E}[W \mid Y = \boldsymbol{y}] = \Sigma_{WY}\Sigma_{YY}^{-1}\boldsymbol{y}.$$

Using the results above:

$$\mathbb{E}[W \mid Y = \boldsymbol{y}] = X^\top(XX^\top + \lambda I)^{-1}\boldsymbol{y}.$$

This is one standard form of the ridge solution:

$$\hat{\boldsymbol{w}} = X^\top(XX^\top + \lambda I)^{-1}\boldsymbol{y}.$$

There is also the equivalent form

$$\hat{\boldsymbol{w}} = (X^\top X + \lambda I)^{-1}X^\top\boldsymbol{y},$$

and the two are related by the matrix identity

$$X^\top(XX^\top + \lambda I)^{-1} = (X^\top X + \lambda I)^{-1}X^\top.$$

9.3 Interpretation

(a) The posterior $p(W \mid Y = \boldsymbol{y})$ is Gaussian, and its mean (also the MAP estimator) is exactly the ridge solution. Minimizing the negative log posterior is equivalent to minimizing squared error plus an ℓ_2 penalty $\lambda\|\boldsymbol{w}\|_2^2$.

(b) Larger λ corresponds to stronger prior/greater noise variance in the observation model. As λ increases, the estimator trusts the prior more and shrinks $\boldsymbol{w}$ more strongly toward 0; as λ decreases, it trusts the data more and approaches the unregularized least-squares estimate.